

From traditional animal breeding to genomic selection

Genomics for Animal Sciences - Summer Workshop (Track 3)

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Nanye Long, Ph.D.

Research Consultant

Institute for Cyber-Enabled Research

Michigan State University

Animal breeding: estimating genetic merit (breeding value)

- Challenge: observed performance is influenced by both genetics and non-genetic factors
- Key idea: separate genetic effects from non-genetic effects by analyzing all available information simultaneously
 - Animal's own performance records
 - Performance of relatives
 - Pedigree (or genomic) relationships
 - Environmental factors
- Outcome: estimated breeding values that can be fairly compared across animals raised in different environments

Henderson's general linear mixed model

- *Applications of Linear Models in Animal Breeding*, by [Charles R. Henderson](#) (1984) -- bridged the gap between theoretical statistics and practical animal breeding
- Established that Best Linear Unbiased Prediction (**BLUP**) yields unbiased predictions of breeding values (BVs) for animals **with and without phenotypic records**
 - Using a numerator/additive relationship matrix (**A**) to capture genetic connections between **all** individuals (key to finding BVs for unphenotyped animals)
 - Using Restricted Maximum Likelihood (REML) to estimate variance components
 - Solving mixed model equations to estimate fixed effects and BVs simultaneously
- Still being used today: **pedigree-based A** → **genomic marker-based G**

The general linear mixed model

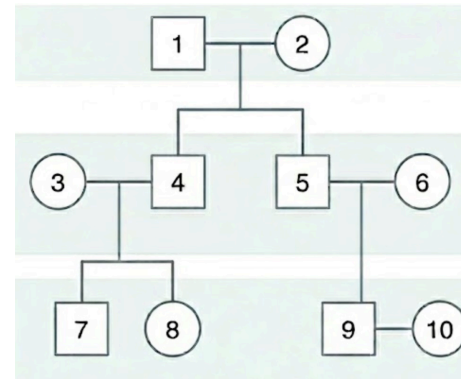
$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \mathbf{e}$$

- $\mathbf{y}$: vector of phenotypic records
- $\boldsymbol{\beta}$: vector of fixed effects (e.g., year, season, farm, sex)
- $\mathbf{u}$: vector of breeding values (random genetic effects)
- $\mathbf{e}$: vector of random residual errors
- $\mathbf{X}, \mathbf{Z}$: design matrices relating the observations to the fixed and random effects

The variances for the random terms are:

- $\text{var}(\mathbf{u}) = \mathbf{A}\sigma_a^2$
- $\text{var}(\mathbf{e}) = \mathbf{I}\sigma_e^2$

Example of the $\mathbf{A}$ matrix

[illegible]

The mixed model equations (MME)

$$\begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{X} & \mathbf{Z}'\mathbf{Z} + \mathbf{A}^{-1}\lambda \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{y} \\ \mathbf{Z}'\mathbf{y} \end{bmatrix}$$

where λ is the ratio of the variance components: $\lambda = \sigma_e^2 / \sigma_a^2$.

- Before constructing the MME, variance components need to be estimated via REML.
- BLUP for $\hat{\mathbf{u}}$ (breeding values for every animal in the analysis) can be solved:

$$\hat{\mathbf{u}} = (\mathbf{Z}'\mathbf{Z} + \mathbf{A}^{-1}\lambda)^{-1}\mathbf{Z}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})$$

These are estimated breeding values (EBV).

- BLUP is a shrinkage estimator: the EBV for an animal with no data is shrunk toward the mean of its relatives.

Toy example of estimating BVs for four animals

- Purpose: illustrate the **flow of genetic information via $\mathbf{A}$** (without the algebraic clutter of the fixed effect adjustments)
- Simplification -- no fixed effects (no intercept)
 - valid only when **phenotypes are not raw observations, but pre-centered**
 - not a full mixed model but a random effects model
- In real-world applications, the model should always include fixed effects.

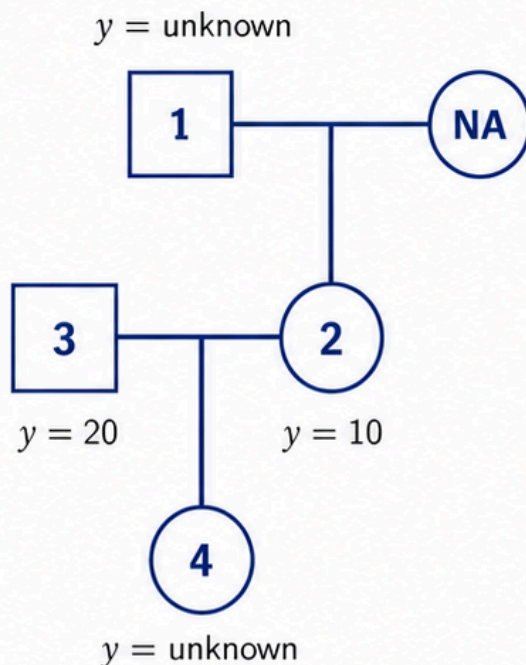
1 Data setup

We have four animals (1, 2, 3, 4):

- **Animal 1:** Founder (un-phenotyped).
- **Animal 2:** Progeny of 1 and an unknown founder (phenotyped).
- **Animal 3:** Founder (phenotyped, unrelated to 1 or 2).
- **Animal 4:** Progeny of 2 and 3 (un-phenotyped).

The Data:

- **Phenotypes (y):** Only animals 2 and 3 have records.
 - $y_2 = 10$
 - $y_3 = 20$
- **Variance Components:**
 - Additive genetic variance ($\sigma_a^2 = 20$)
 - Residual variance ($\sigma_e^2 = 40$)
 - $\lambda = \frac{\sigma_e^2}{\sigma_a^2} = \frac{40}{20} = 2$



5 Insight

1. **Direct Shrinkage:** Animals 2 and 3 have their phenotypes “shrunk” by the denominator $(1 + \lambda)$. Since $\lambda = 2$, they keep exactly $1/3$ of their performance as their genetic merit.
2. **Parental Inheritance:** Animal 1 (unphenotyped) gets exactly $5/3$, which is half of its offspring's $10/3$. This demonstrates the $1/2$ genetic flow from parent to child.
3. **Progeny Prediction:** Animal 4 (unphenotyped) gets exactly 5, which is the mid-parent average: $(10/3 + 20/3)/2 = 30/6 = 5$.

2 Forming vector and matrices

$$y = \begin{bmatrix} 10 \\ 20 \end{bmatrix} \quad Z = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1/2 & 0 & 1/4 \\ 1/2 & 1 & 0 & 1/2 \\ 0 & 0 & 1 & 1/2 \\ 1/4 & 1/2 & 1/2 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 4/3 & -2/3 & 0 & 0 \\ -2/3 & 11/6 & 1/2 & -1 \\ 0 & 1/2 & 3/2 & -1 \\ 0 & -1 & -1 & 2 \end{bmatrix}$$

3 Solve MME

$$\begin{bmatrix} X'X & X'Z \\ Z'X & Z'Z + A^{-1}\lambda \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{u} \end{bmatrix} = \begin{bmatrix} X'y \\ Z'y \end{bmatrix}$$

where $\lambda = 2$ and A is the numerator relationship matrix.

4 Final predicted BVs

$$\hat{u} = \begin{bmatrix} 5/3 \\ 10/3 \\ 20/3 \\ 5 \end{bmatrix} \quad \begin{array}{l} \text{Animal 1} \\ \text{Animal 2} \\ \text{Animal 3} \\ \text{Animal 4} \end{array}$$

R code for the toy example

```
# --- 0. Clear Environment ---
rm(list = ls())

# --- 1. Pedigree Matrix (A) - 4x4 ---
# Rows represent animals 1, 2, 3, 4.
A <- matrix(c(1.0, 0.5, 0.0, 0.25,
              0.5, 1.0, 0.0, 0.5,
              0.0, 0.0, 1.0, 0.5,
              0.25, 0.5, 0.5, 1.0),
            nrow = 4, byrow = TRUE)

# --- 2. Design Matrix (Z) - 2x4 ---
# Explicitly defines 2 rows (phenotypes) and 4 columns (animals).
# Animal 2 gets Record 1; Animal 3 gets Record 2.
Z <- matrix(c(0, 1, 0, 0, # Row 1: Phenotype for Animal 2
              0, 0, 1, 0), # Row 2: Phenotype for Animal 3
            nrow = 2, byrow = TRUE)

# --- 3. Phenotypes (y) and k ---
y <- matrix(c(10, 20), nrow = 2) # 2x1 Column Matrix
k <- 2 # k = lambda in the diagram

# --- 4. The MME Setup ---
# Henderson's Mixed Model Equations: (Z'Z + A^-1 * k) * u = Z'y
A_inv <- solve(A)
ZZ <- t(Z) %*% Z # Result: 4x4
Zy <- t(Z) %*% y # Result: 4x1

# --- 5. Solving for Breeding Values (u_hat) ---
LHS <- ZZ + (A_inv * k)
u_hat <- solve(LHS, Zy)

# --- 6. Results ---
cat("--- MME Solutions (u_hat) ---\n")
print(as.vector(u_hat))
```

With genomic data

- We use **G** (genomic relationship matrix) instead of the traditional pedigree **A** matrix
 - BLUP $\Rightarrow$ GBLUP, and EBV $\Rightarrow$ GEBV
- Comparison
 - **A** (expectation): e.g., full siblings share exactly 50% of their alleles.
 - **G** (realization): captures the actual Mendelian sampling. Some full-sibs might share 42% and others 58%.

How to build **G**?

$$\text{Raw genotype} = \begin{bmatrix} 1 & 0 & -1 & 1 & 0 \\ -1 & 1 & 0 & 0 & -1 \\ 0 & -1 & 1 & -1 & 1 \end{bmatrix}$$

(three animals on five SNP markers; coded by aa=-1, Aa=0, AA=1)

Let **M** be the **column-centered** matrix of the above raw genotypes,

$$\mathbf{G} = \mathbf{M}\mathbf{M}' / \sum_{i=1}^5 2p_iq_i$$

$\sum_{i=1}^5 2p_iq_i$ is the sum of the variances of all markers. This scales **G** so that the average diagonal element is approximately 1, making it comparable to the pedigree **A** matrix. ([VanRaden, 2008, J. Dairy Sci.](#))

The **rrBLUP** R package

- User manual: <https://jendelman.r-universe.dev/rrBLUP/doc/manual.html>
- Implements two **equivalent** models in a single function `mixed.solve()`, depending on your input parameter:
 - i. marker perspective ("RR-BLUP" or "SNP-BLUP")
 - You provide **SNP genotype matrix** -- model estimates effect of each SNP. You sum them up to get the animal's EBV.
 - ii. animal perspective ("G-BLUP")
 - You provide **genomic relationship matrix** -- model estimates EBV of each animal.

Code demo using rrBLUP

[Access the "rrBLUP" code](#)

- Simulated 100 animals (split into training set and validation set), 500 markers, i.e., the "small n , large p " problem
- Simulated a heritability of 0.45
- Under "small n , large p " and low/moderate heritability, the model tends to fit the noise, i.e., **overfitting**
 - Artificially inflated training accuracy
 - Low and highly fluctuating validation accuracy
 - Inaccurate and fluctuating heritability estimate
 - **Solution:** feeding the model with more phenotypes

```

--- Heritability validation ---
TRUE Simulated h2 : 0.450000
Estimated G-BLUP : 0.247283
Estimated RR-BLUP : 0.247283

```

Discrepancy b/w true h2 and estimated h2, due to overfitting

```

--- Prediction equivalence on training animals (first 20) ---

```

Animal	True_BV	GBLUP_EBV	RRBLUP_EBV
1 Animal_1	4.4	1.93574248	1.93573998
2 Animal_2	14.5	10.60368019	10.60366782
3 Animal_3	10.7	0.98872971	0.98872784
4 Animal_4	1.4	3.62345449	3.62345105
5 Animal_5	-0.4	3.42570395	3.42569974
6 Animal_6	-4.7	1.09661757	1.09661643
7 Animal_7	8.2	7.38760250	7.38759385
8 Animal_8	-9.4	-7.32802126	-7.32801350
9 Animal_9	3.6	1.00603188	1.00603021
10 Animal_10	-4.6	-3.56343140	-3.56342673
11 Animal_11	-7.1	-2.20900317	-2.20900077
12 Animal_12	-4.0	-3.90671687	-3.90671340
13 Animal_13	-7.6	0.14828603	0.14828502
14 Animal_14	-0.1	0.51663498	0.51663426
15 Animal_15	-1.6	-0.06444747	-0.06444832
16 Animal_16	-10.1	-3.21654525	-3.21654110
17 Animal_17	-11.7	-3.12613131	-3.12612765
18 Animal_18	-5.8	-0.28498575	-0.28498545
19 Animal_19	4.7	0.84124039	0.84123981
20 Animal_20	0.5	-3.26210947	-3.26210465

GBLUP and RR-BLUP produced the same predictions on both training (with phenotypes) and validation (no phenotypes) sets

```

--- Prediction equivalence on validation animals ---

```

Animal	True_BV	GBLUP_EBV	RRBLUP_EBV
1 Animal_81	-4.2	0.58370189	0.58370180
2 Animal_82	-1.6	-0.61101812	-0.61101727
3 Animal_83	-1.5	-0.07728269	-0.07728259
4 Animal_84	-1.2	-1.87344517	-1.87344275
5 Animal_85	-24.2	-1.22187943	-1.22187773
6 Animal_86	3.8	-0.18160803	-0.18160804
7 Animal_87	-0.4	1.45981267	1.45981066
8 Animal_88	-13.0	0.36946894	0.36946849
9 Animal_89	5.1	-0.45512092	-0.45512015
10 Animal_90	-6.2	-1.15924397	-1.15924259
11 Animal_91	5.9	-2.31187459	-2.31187151
12 Animal_92	-6.1	-0.58482067	-0.58482021
13 Animal_93	-6.1	0.11050862	0.11050874
14 Animal_94	9.9	0.13077819	0.13077784
15 Animal_95	17.4	3.53632135	3.53631723
16 Animal_96	-5.7	-0.27242672	-0.27242653
17 Animal_97	-4.9	-1.31334300	-1.31334160
18 Animal_98	-4.6	0.39689202	0.39689144
19 Animal_99	1.5	-1.87190988	-1.87190764
20 Animal_100	-7.2	1.79778300	1.79778086

High training accuracy and low validation accuracy, due to overfitting. Also, with only 20 animals for validation, the result fluctuates in each run.

```

--- Accuracy (correlation b/w TBV and EBV) ---

```

```

Training data accuracy: 0.742140
Validation data accuracy: 0.284866

```

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